Supplementary Material: Spatiotemporal Aliasing in Video Action Recognition: A Cross-Architecture Analysis at Scale

Anonymous ACCV 2026 Submission

Paper ID #*****

This document provides supplementary material for the main paper. We include: (S1) full coverage×stride heatmaps for all 8 models and 8 datasets; (S2) Levene variance inflation analysis; (S3) full ANOVA effect size tables including FineGym; (S4) action sensitivity taxonomy detail; (S5) entropy routing curves for all models; (S6) clip duration analysis; (S7) spectral correlation detail; (S8) implementation details (target-resolution fine-tuning, positional embedding interpolation); (S9) full multi-dataset spatial resolution results (all 8 datasets $\times$ 8 models); (S10) interactive dashboard and code availability; and (S11) leave-one-architecture-out stability analysis of the TDS ranking.

S1 Full Coverage×Stride Heatmaps

Figures S1–S8 show the complete coverage×stride accuracy heatmaps for all 8 architectures across all 8 datasets. Each cell represents Top-1 accuracy (%) at a specific (coverage, stride) configuration. Color scale: green=high accuracy, red=low accuracy. FineGym heatmaps (all 8 models) are included; per-class optical-flow analysis for FineGym is reported in supplementary S7.

Row order (coverage): 10%, 25%, 50%, 75%, 100%. Column order (stride): 1, 2, 4, 8, 16.

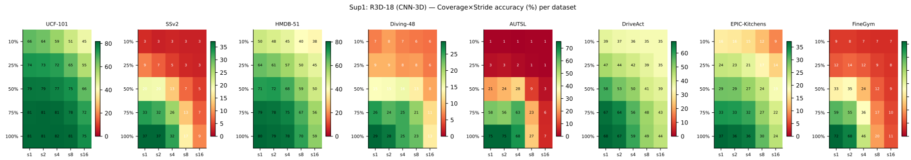


Fig. S1: R3D-18 (CNN-3D) coverage×stride accuracy (%) across 8 datasets. Sharp cliff at stride 4→8 on AUTSL (75%→27%) and similar cliff on FineGym confirms the high temporal demand of fine-grained actions. EPIC-Kitchens and UCF-101 retain high accuracy even at sparse sampling.

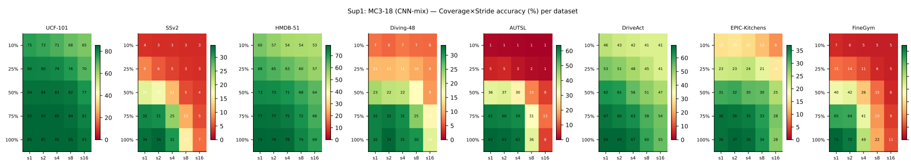


Fig. S2: MC3-18 (CNN-mix, channel-separated 3D) heatmaps. Similar pattern to R3D-18 but with slightly reduced aliasing on low-TDS datasets (UCF-101, DriveAct), consistent with the regularizing effect of 2D-channel separation. FineGym shows the same severe cliff pattern as AUTSL.

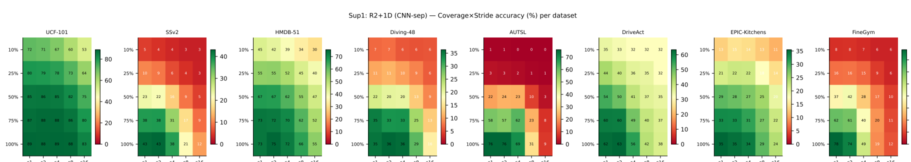


Fig. S3: R2+1D-18 (separable spatiotemporal convolution) heatmaps. Higher base accuracy than R3D-18 on several datasets (SSv2: 42.6% vs 37.1%), but similar aliasing sensitivity pattern—cliff at stride 4→8 on AUTSL and FineGym.

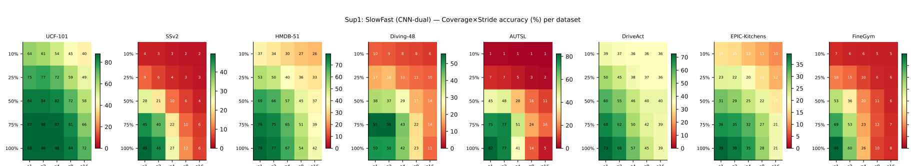


Fig. S4: SlowFast-R50 (dual-pathway) heatmaps. Cliff occurs at stride 2→4 on AUTSL (77%→41%)—one step earlier than other CNNs—because the fast pathway requires 32 frames. On FineGym, SlowFast achieves the highest base accuracy (78.2%) among all models but suffers the worst aliasing loss (71.5pp at stride=16).

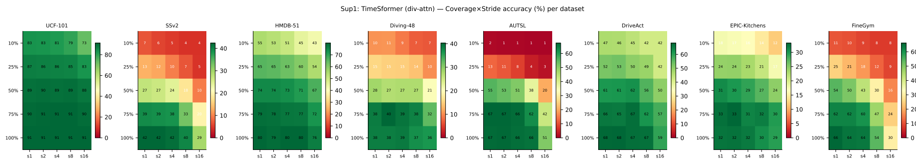


Fig. S5: TimeSformer (divided space–time attention) heatmaps. Notably flat across most stride values, with steep degradation only at the combined extreme of low coverage *and* high stride. FineGym exception: drops 36.1pp at stride=16, revealing that even global divided attention cannot fully compensate for the 6–8Hz gymnastics phase transitions.

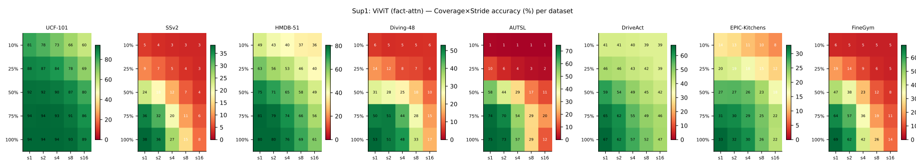


Fig. S6: ViViT (factorized attention) heatmaps. Despite being a Transformer, ViViT aliases nearly as severely as 3D CNNs (AUTSL: cliff at stride 2→4, identical to SlowFast). FineGym shows a 55.2pp drop at stride=16, confirming that factorized temporal attention leaves the model vulnerable to frame dropout in high-TDS domains.

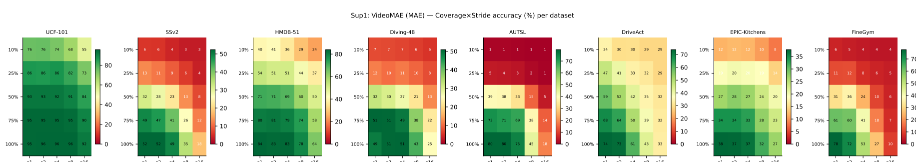


Fig. S7: VideoMAE (masked autoencoder) heatmaps. High base accuracy (SSv2: 52.4%, UCF-101: 95.4%, FineGym: 78.0%) but steep temporal degradation (AUTSL cliff at stride 4→8; FineGym: 67.8pp drop). The MAE pretraining objective is spatial—reconstructing patches—and does not transfer to temporal robustness.

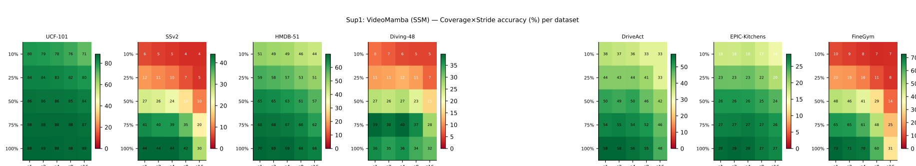


Fig. S8: VideoMamba (SSM selective scan) heatmaps (all 8 datasets). On 5 of 8 datasets, VideoMamba shows a flat heatmap with minimal degradation across strides—the SSM recurrent state aggregates temporal context, providing natural robustness. FineGym (40.8pp), AUTSL (16.0pp), and SSv2 (14.1pp) are exceptions, revealing the limit of SSM temporal integration under high-frequency transitions.

S2 Levene Variance Inflation Analysis

Figure S9 plots inter-class standard deviation at stride=1 (x-axis) versus stride=16 (y-axis) for all model-dataset pairs. Points above the diagonal indicate that sparse sampling increases inter-class variability. Significant pairs ($p < 0.05$, Bonferroni-corrected) are highlighted in red.

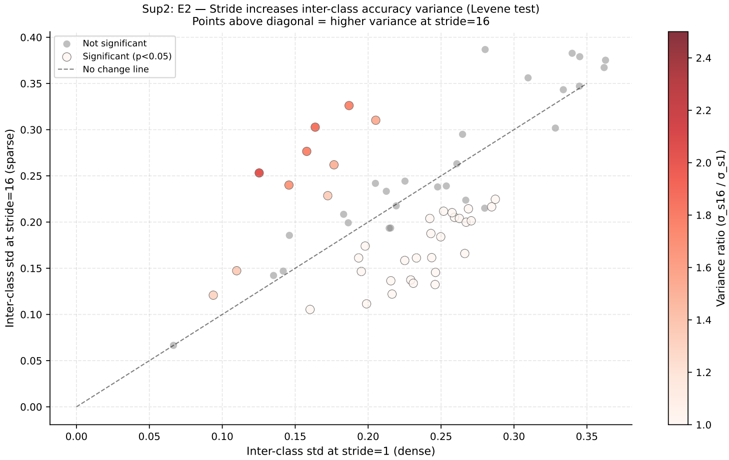


Fig. S9: Levene test: inter-class accuracy std at stride=1 vs. stride=16. Points above the dashed diagonal have higher variance under sparse sampling. Color encodes the variance ratio (σ_{s16}/σ_{s1}). **Key finding**: 67% of model-dataset pairs show significant variance inflation ($p < 0.05$), with the largest increases in VideoMAE/HMDB-51 (2.0×) and R3D-18/HMDB-51 (1.85×). TimeSformer and VideoMamba show stable variance (≈1.1×).

Table S1 lists the 10 model-dataset pairs with highest variance inflation, quantifying the instability masked by mean accuracy.

Table S1: Top-10 variance inflation cases (stride=1→16, coverage=100%). All rows significant at p<0.05.

Model	Dataset	$\sigma_{s=1}$	$\sigma_{s=16}$	Ratio	Levene p
VideoMAE	HMDB-51	0.125	0.253	2.02×	<0.0001
R3D-18	HMDB-51	0.164	0.303	1.85×	0.0001
ViViT	HMDB-51	0.158	0.277	1.75×	0.0004
SlowFast	HMDB-51	0.187	0.326	1.74×	<0.0001
SlowFast	UCF-101	0.146	0.240	1.65×	<0.0001
R2+1D	HMDB-51	0.205	0.310	1.51×	0.0021
MC3-18	HMDB-51	0.177	0.262	1.48×	0.0187
ViViT	UCF-101	0.110	0.147	1.34×	0.0093
VideoMAE	AUTSL	0.172	0.229	1.33×	0.0324
VideoMAE	UCF-101	0.094	0.121	1.29×	0.0239

S3 Full ANOVA Effect Size Tables

Figure S10 shows the mean η^2 per model (averaged over all valid datasets), with error bars indicating standard deviation.

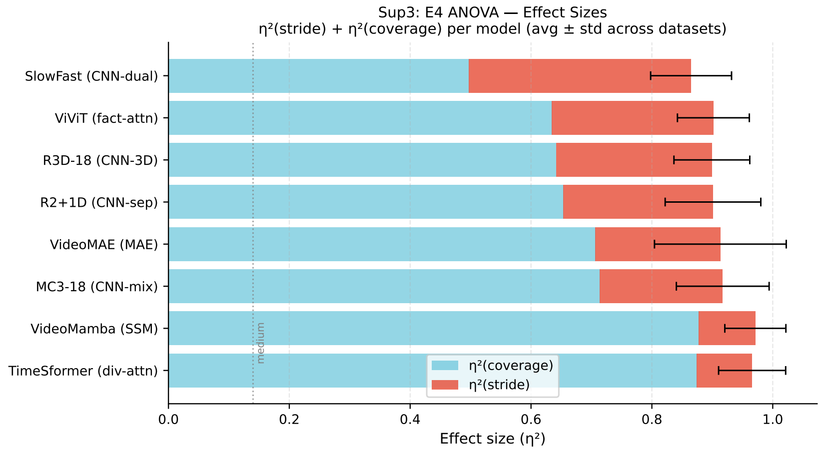


Fig. S10: ANOVA effect sizes: η^2(coverage) + η^2(stride) per model. Coverage consistently dominates (blue bars, 53–90% of variance). The stride effect is large for CNNs and ViViT (red bars, 18–35%) but small for VideoMamba and TimeSformer (<12%), confirming architecture-dependent aliasing. Error bars: std across all 8 datasets.

Table S2 provides the complete per-dataset η^2 values for all 8 datasets. All stride effects are significant ($p < 0.05$); coverage effects are significant ($p < 0.001$) in all cases. FineGym (η^2_{stride} up to 0.50 for SlowFast) and AUTSL show the largest stride effects of all 8 datasets, consistent with their co-equal top ranking on TDS (FineGym=55.9pp, AUTSL=53.0pp; see main paper Sec. 3.1 and supplementary S11 for the leave-one-out sensitivity of their relative order).

Table S2: Full ANOVA η^2 per model and dataset. Cov = η^2(coverage), Str = η^2(stride). Rows shaded darker indicate larger stride effect. FineGym column added with values computed from the full coverage×stride sweep.

Model	AUTSL		FineGym		SSv2		Diving-48		HMDB-51		DriveAct		EPIC		UCF-101	
	Cov	Str	Cov	Str	Cov	Str	Cov	Str	Cov	Str	Cov	Str	Cov	Str	Cov	Str
VMamba	.88	.09	.77	.16	.88	.09	.94	.04	.90	.09	.83	.15	.95	.05	.88	.07
TSF	.92	.06	.74	.19	.87	.10	.97	.02	.92	.07	.84	.14	.91	.08	.85	.07
MC3	.66	.18	.50	.33	.54	.28	.81	.13	.81	.18	.75	.22	.89	.11	.79	.18
R3D	.60	.21	.47	.34	.60	.26	.77	.15	.68	.31	.58	.33	.75	.24	.70	.25
R2+1D	.63	.20	.50	.32	.62	.26	.76	.17	.72	.27	.49	.38	.80	.18	.76	.20
VMAE	.69	.18	.53	.28	.69	.22	.82	.12	.84	.16	.43	.43	.87	.11	.81	.14
ViViT	.61	.24	.53	.32	.51	.33	.69	.20	.71	.27	.60	.31	.80	.19	.68	.26
SF	.51	.29	.26	.50	.44	.36	.51	.33	.55	.39	.39	.46	.67	.30	.61	.36

S4 Action Sensitivity Taxonomy

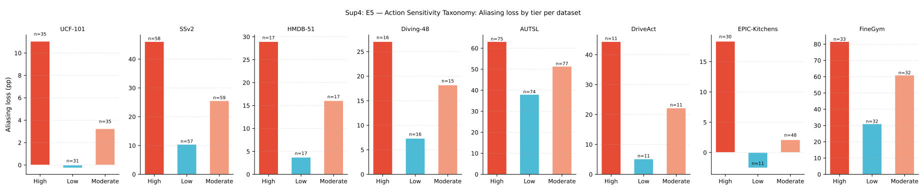


Fig. S11: Aliasing loss (pp) per sensitivity tier and dataset. Each bar shows mean accuracy drop (stride=1→16) within the tier, annotated with the number of action classes. UCF-101 Low tier: −0.3pp (static, appearance-dominated actions lose essentially no accuracy under sparse sampling). AUTSL: all three tiers remain high (> 38pp), confirming that sign language is fundamentally high-frequency regardless of action subclass.

Table S3 lists the threshold values that define tier boundaries per dataset, derived from the 33rd and 67th percentiles of the per-class accuracy drop.

Table S3: Taxonomy tier thresholds (accuracy drop, pp): classes below the Low threshold alias negligibly; classes above the High threshold are severely affected. Thresholds are dataset-specific and computed across all valid architectures.

Dataset	Classes	Low threshold	High threshold	High count	Low count
AUTSL	226	66.9pp	73.9pp	75	74
Diving-48	47	25.0pp	47.7pp	16	16
SSv2	174	59.7pp	76.9pp	58	57
HMDB-51	51	8.4pp	28.5pp	17	17
EPIC-Kitchens	89	0.0pp	19.4pp	30	11
DriveAct	33	14.8pp	51.0pp	11	11
UCF-101	101	1.4pp	7.5pp	35	31

S5 Entropy Routing Curves — All Models

Figure S12 shows accuracy vs. average frames for all 8 architectures on SSv2 under the entropy routing protocol. Each curve is the result of sweeping confidence threshold $\tau \in [0.10, 0.95]$. Dashed horizontal lines indicate fixed-budget baselines (4f, 16f) and the oracle.

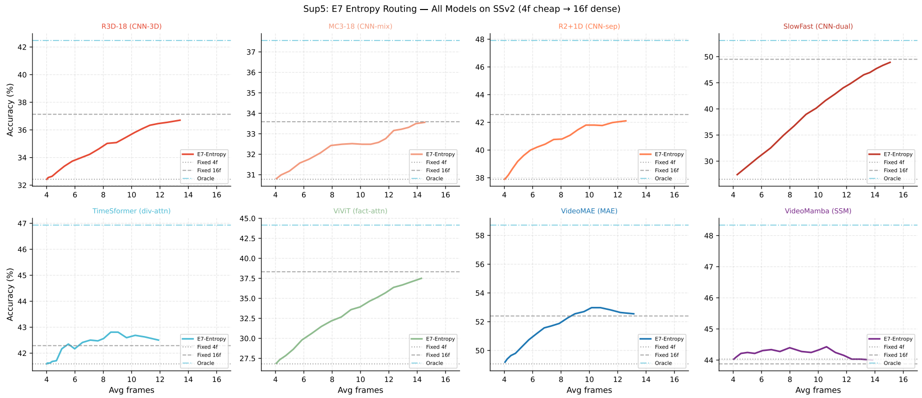


Fig. S12: Entropy routing: accuracy vs. avg frames (SSv2, 4f cheap → 16f dense). For architectures with flat 4f≈16f accuracy (TimeSformer, VideoMamba), routing near-trivially selects 4f for most videos and matches dense accuracy. For architectures with large 4f–16f gaps (SlowFast, ViViT), the routing curve provides meaningful operating points between the fixed-budget extremes.

Table S4 provides the full entropy routing results across all models and datasets at the optimal threshold (avg frames ≤ 8).

Table S4: Entropy routing: best accuracy at avg ≤ 8 frames per model/dataset. Values are Top-1 accuracy (%). Bold = highest per column. *VideoMamba/AUTSL: entropy routing not computed; value is direct stride=8 accuracy (65.0%).

Model	AUTSL	FineGym	Diving	SSv2	HMDB	DriveAct	EPIC	UCF	Avg
R3D-18	73.3	68.5	27.6	34.6	80.2	66.1	36.7	81.6	58.6
MC3-18	63.4	62.9	31.8	32.4	78.5	69.4	36.5	85.6	57.6
R2+1D	74.2	72.9	36.3	40.8	73.3	59.6	34.8	89.0	60.1
SlowFast	59.5	57.8	46.6	35.0	76.3	67.4	37.4	87.9	58.5
TimeSformer	67.1	66.8	38.9	42.5	80.6	68.5	32.4	91.3	**61.0**
ViViT	67.2	66.1	48.9	32.2	80.1	64.3	31.3	94.3	60.6
VideoMAE	**79.5**	**76.4**	**52.6**	**51.9**	**84.7**	**69.0**	**37.8**	**95.9**	**68.5**
VideoMamba	65.0*	73.2	37.4	44.4	69.8	57.6	28.4	88.5	58.0

S6 Clip Duration Analysis

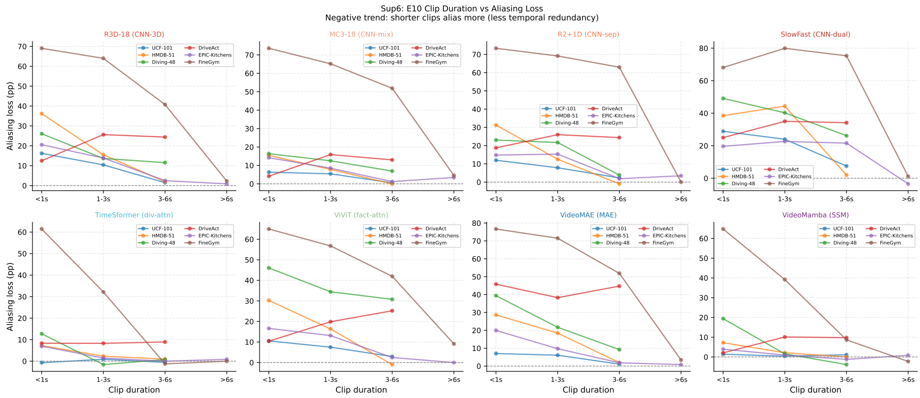


Fig. S13: Aliasing loss (pp) vs. clip duration bin for all 8 architectures across multiple datasets. Counter-intuitive finding: **shorter clips alias more** (Pearson $r = -0.3$ to -0.8 across models). Short clips have less temporal redundancy; any frame drop removes a larger fraction of the available evidence. Long clips (>4s) contain natural temporal repetition that compensates for sparse sampling. Exception: Diving-48 longer clips (>4s) alias more, consistent with longer dives containing more distinct phase transitions.

S7 Spectral Correlation Detail

We estimated per-class temporal frequency demand using optical-flow magnitude (Farneback algorithm [1]) computed on 5 representative videos per class, sampling 16 frames per video at 112×112px. The mean inter-frame flow magnitude serves as a proxy for temporal activity level. This analysis is exploratory and should not be read as confirmatory evidence for the Nyquist-aliasing hypothesis on its own; see the main paper (Sec. 3.1, Fig. 3 right) for the summary figure including FineGym.

Table S5 reports Pearson correlation between per-class mean flow magnitude and aliasing loss (accuracy drop stride=1→16) per dataset. Prior studies on small closed-vocabulary datasets (UCF-101, Kinetics-400) have reported near-perfect optical-flow–aliasing correlations (r=0.99) using controlled synthetic flow protocols. Our large-scale evaluation on 8 diverse datasets yields a more nuanced picture ($r = -0.475$ to 0.327). This has at least four plausible explanations: (1) complex background motion in egocentric datasets (EPIC-Kitchens) decorrelates flow from action content; (2) sign language (AUTSL) involves localized, subtle movements that optical flow at 112px partially misses; (3) our use of raw flow magnitude rather than class-conditioned frequency analysis is a coarser proxy; (4) FineGym's negative correlation suggests that for discrete-event classes (a held position, a precise landing), raw motion magnitude is the wrong signal entirely—what matters is whether sparse sampling catches the one brief moment that defines the class, which low-flow (slow, precise) actions are more likely to lose than high-flow (fast, sustained) ones.

Table S5: Spectral correlation: Pearson r between optical-flow magnitude and per-class aliasing loss. Positive r supports the Nyquist hypothesis; near-zero or negative values suggest flow magnitude is an insufficient (or inverted) proxy for temporal demand.

Dataset	Pearson r	p-value	Significant (p<0.05)
DriveAct	0.327	0.065	Marginal
AUTSL	0.285	<0.001	Yes
SSv2	0.184	<0.001	Yes
HMDB-51	0.180	0.206	No
Diving-48	0.140	0.349	No
UCF-101	0.031	0.762	No
EPIC-Kitchens	–0.002	0.985	No
FineGym	–0.475	<0.0001	Yes (inverted)

S8 Implementation Details

Hardware. Training experiments used NVIDIA A100 (40GB) and H200 NVL (141GB) GPUs (CUDA 13.2). Temporal-sweep evaluation (1,600 configurations) was parallelized across 8 nodes using a custom Slurm auto-submitter. Inference latency measurements used an NVIDIA RTX PRO 6000 Blackwell GPU (96GB) with bfloat16 autocast, batch size=1, 20 warmup iterations, 100 benchmark iterations, and CUDA event timing with explicit synchronization.

Training hyperparameters. All models fine-tuned for 10 epochs from Kinetics-400 checkpoints. Transformers and VideoMamba: AdamW, lr=2×10^{-5}, weight decay=0.05, batch size=32 per GPU. CNNs: AdamW, lr=10^{-4}, batch size=128 per GPU. SlowFast: AdamW, lr=10^{-4}, batch size=32 per GPU. All models use cosine learning rate decay with linear warmup (1 epoch). Spatial augmentation: random resized crop, horizontal flip, normalization (ImageNet statistics).

Positional embedding interpolation for Transformers and SSMs. For target-resolution fine-tuning at non-native resolutions, Transformer and SSM positional embeddings must be adapted. *The wrong approach*: using the HuggingFace mismatched-size flag discards positional embeddings entirely, causing a large accuracy cliff (typically 10–20pp worse than our approach). *Our approach*: load the native-resolution checkpoint, then bicubically interpolate spatial positional embeddings from the native grid (14×14=196 tokens at 224px) to the target grid ($\lfloor r/16 \rfloor^2$ tokens at resolution r) before copying weights to a model instantiated at the target resolution. This preserves the spatial topology learned during pre-training and ensures fine-tuning starts from a meaningful positional encoding rather than a random one. The same bicubic interpolation applies to TimeSformer, VideoMAE, ViViT, and VideoMamba.

SlowFast spatial pooling fix. The original SlowFast-R50 head uses a fixed 7×7 average pooling kernel, assuming 224px input. For non-native resolutions, we replace both head pooling layers with `AdaptiveAvgPool3d(1)`, enabling forward passes at arbitrary input resolutions without parameter changes.

FineGym dataset details. FineGym [2] contains 97 fine-grained gymnastics action classes (floor exercise, uneven bars, balance beam, vault; parallel bars, horizontal bar, pommel horse, rings for men). We use the FineGym288 subset (1,960 clips), balanced to 20 clips per class for training and evaluation. Preprocessing: clips are trimmed to the annotated action boundary (mean duration 3.2s at 25fps) and padded with the last frame if shorter than 16 frames.

S9 Multi-Dataset Spatial Resolution Results

Tables S6–S13 extend the spatial robustness analysis from the main paper (Section 4.3) to all 8 datasets. Tables S6–S12 show cross-resolution results (no retraining): the fine-tuned native-resolution checkpoint evaluated at 96–336px. Table S13 shows FineGym target-resolution fine-tuning results (one fine-tuned checkpoint per resolution). **Bold** = native training resolution. $\Delta_{\max}$ = largest accuracy drop from native across all tested resolutions.

Table S6: Spatial robustness on **SSv2** (repeated from main paper; no retraining).

Model	Native	96px	112px	160px	224px	336px	$\Delta_{\max}$
R3D-18	112px	35.9	**37.1**	30.3	17.2	5.8	−31.4pp
MC3-18	112px	32.4	**33.6**	27.8	14.8	4.4	−29.2pp
R2+1D	112px	40.1	**42.6**	36.2	20.8	5.9	−36.6pp
SlowFast	224px	27.7	32.4	45.7	**49.5**	36.0	−21.8pp
TimeSformer	224px	39.3	39.4	41.4	**42.3**	42.1	−3.0pp
ViViT	224px	36.4	37.1	38.1	**38.3**	38.0	−1.9pp
VideoMAE	224px	48.9	49.2	51.5	**52.3**	51.9	−3.4pp
VideoMamba	224px	37.5	39.3	42.5	**43.9**	43.8	−6.4pp

Table S7: Spatial robustness on **UCF-101** (no retraining).

Model	Native	96px	112px	160px	224px	336px	$\Delta_{\max}$
R3D-18	112px	79.9	**81.2**	75.3	55.1	20.7	−60.5pp
MC3-18	112px	82.9	**85.4**	81.0	63.9	24.9	−60.5pp
R2+1D	112px	88.0	**88.6**	85.1	64.9	25.0	−63.6pp
SlowFast	224px	69.7	75.8	87.5	**91.4**	89.2	−21.7pp
TimeSformer	224px	89.1	89.7	90.7	**91.5**	90.9	−2.4pp
ViViT	224px	92.0	93.5	94.7	**95.1**	95.0	−3.1pp
VideoMAE	224px	95.3	95.7	96.1	**96.4**	96.3	−1.1pp
VideoMamba	224px	76.7	78.7	82.6	**83.5**	83.8	−6.7pp

Table S8: Spatial robustness on **HMDB-51** (no retraining).

Model	Native	96px	112px	160px	224px	336px	$\Delta_{\max}$
R3D-18	112px	78.8	**80.3**	75.9	59.8	24.7	−55.6pp
MC3-18	112px	76.7	**78.6**	74.2	51.0	14.5	−64.2pp
R2+1D	112px	74.4	**73.1**	68.4	50.5	16.2	−56.9pp
SlowFast	224px	59.5	67.2	80.5	**81.6**	79.0	−22.0pp
TimeSformer	224px	75.9	77.2	77.5	**78.9**	78.0	−2.9pp
ViViT	224px	78.0	79.0	79.9	**79.5**	80.0	−1.6pp
VideoMAE	224px	83.7	83.1	84.0	**83.8**	84.1	−0.8pp
VideoMamba	224px	59.9	63.8	67.2	**69.8**	68.9	−9.9pp

Table S9: Spatial robustness on **Diving-48** (no retraining).

Model	Native	96px	112px	160px	224px	336px	$\Delta_{\max}$
R3D-18	112px	26.5	**28.8**	22.1	12.8	7.3	−21.5pp
MC3-18	112px	30.8	**31.6**	27.3	13.7	7.3	−24.4pp
R2+1D	112px	33.8	**35.3**	30.1	18.7	7.8	−27.5pp
SlowFast	224px	18.4	25.0	43.3	**50.5**	41.6	−32.0pp
TimeSformer	224px	37.4	39.3	38.6	**38.1**	38.6	−0.7pp
ViViT	224px	48.0	49.9	52.8	**53.0**	51.3	−5.0pp
VideoMAE	224px	48.7	48.3	50.9	**48.6**	49.1	−0.3pp
VideoMamba	224px	29.2	29.1	36.1	**36.6**	36.2	−7.5pp

Table S10: Spatial robustness on **AUTSL** (sign language; no retraining from AUTSL-fine-tuned checkpoint). CNN degradation is most severe: R3D-18 collapses from 75.0% to 2.0% at 336px (−73.0pp). VideoMamba excluded: K400-pretrained checkpoint not fine-tuned on AUTSL; temporal aliasing results (Table 2) use the AUTSL-fine-tuned checkpoint.

Model	Native	96px	112px	160px	224px	336px	$\Delta_{\max}$
R3D-18	112px	72.2	**75.0**	63.0	27.9	2.0	−73.0pp
MC3-18	112px	59.1	**63.7**	52.7	12.2	1.2	−62.5pp
R2+1D	112px	73.7	**75.9**	68.9	38.4	7.3	−68.7pp
SlowFast	224px	22.5	32.6	70.0	**79.2**	69.0	−56.7pp
TimeSformer	224px	65.1	66.0	68.3	**68.5**	68.6	−3.5pp
ViViT	224px	49.2	51.8	53.3	**53.6**	54.0	−4.5pp
VideoMAE	224px	77.3	78.2	79.5	**79.6**	79.3	−2.2pp
VideoMamba	224px	K400-pretrained checkpoint collapses on AUTSL (domain gap)					

Table S11: Spatial robustness on **DriveAct** (in-cabin activities; no retraining).

Model	Native	96px	112px	160px	224px	336px	$\Delta_{\max}$
R3D-18	112px	65.0	**68.3**	52.7	19.0	4.9	−63.4pp
MC3-18	112px	65.2	**69.0**	38.4	6.2	4.5	−64.5pp
R2+1D	112px	67.4	**62.5**	40.0	12.9	4.5	−58.0pp
SlowFast	224px	31.7	40.2	65.8	**70.1**	58.7	−38.4pp
TimeSformer	224px	68.3	69.2	69.6	**70.3**	69.9	−2.0pp
ViViT	224px	59.4	58.7	62.1	**63.4**	63.6	−4.7pp
VideoMAE	224px	71.4	69.9	71.7	**73.0**	71.4	−3.1pp
VideoMamba	224px	35.0	38.2	46.9	**57.8**	52.9	−22.8pp

Table S12: Spatial robustness on **EPIC-Kitchens** (egocentric cooking; no retraining).

Model	Native	96px	112px	160px	224px	336px	$\Delta_{\max}$
R3D-18	112px	33.3	**37.2**	26.7	8.6	3.0	−34.1pp
MC3-18	112px	36.1	**36.2**	27.2	9.4	2.5	−33.6pp
R2+1D	112px	32.8	**35.5**	27.6	11.8	4.6	−30.9pp
SlowFast	224px	11.3	17.7	31.3	**37.5**	28.7	−26.2pp
TimeSformer	224px	20.5	19.2	25.6	**31.7**	23.5	−12.5pp
ViViT	224px	30.2	34.1	36.1	**37.9**	36.0	−7.7pp
VideoMAE	224px	30.9	32.6	36.5	**38.6**	39.1	−7.7pp
VideoMamba	224px	11.1	12.6	20.8	**28.3**	24.5	−17.3pp

Table S13: FineGym spatial resolution results after target-resolution fine-tuning (one checkpoint per resolution, 10 fine-tuning epochs from native checkpoint). Bold = native training resolution (112px for CNNs, 224px for Transformers/SSM). Unlike other datasets in S9 (cross-resolution without retraining), FineGym shows CNN performance that *improves beyond native* after fine-tuning at higher resolutions (R3D-18: 71.5%@112px→76.1%@224px), confirming that higher spatial resolution genuinely aids gymnastics pose discrimination.

Model	Native	48px	96px	112px	160px	224px
R3D-18	112px	56.4	72.6	**71.5**	74.9	76.1
MC3-18	112px	46.3	62.3	**64.3**	73.8	65.7
R2+1D	112px	61.6	73.1	**76.8**	79.4	76.2
SlowFast	224px	36.3	67.2	69.6	76.3	**78.2**
TimeSformer	224px	24.3	48.9	50.8	62.2	**66.4**
ViViT	224px	21.3	39.1	40.8	49.6	**69.9**
VideoMAE	224px	25.4	51.1	57.5	64.8	**78.0**
VideoMamba	224px	29.1	51.6	58.2	66.0	**73.2**

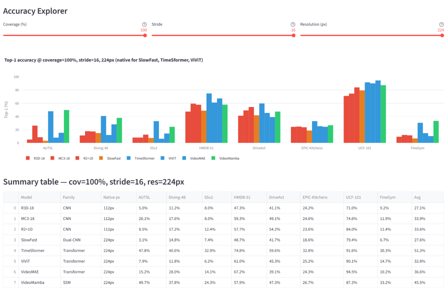
S10 Interactive Dashboard and Code Availability

Interactive visualization dashboard. An anonymized interactive dashboard is included with the supplementary material for review. The public dashboard URL will be released after the review period. The dashboard provides:

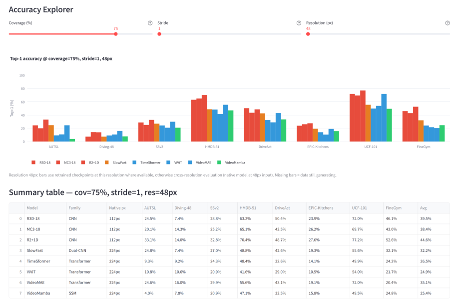
1. **Temporal sweep explorer**: coverage×stride heatmaps per model and dataset, with configurable color scaling and reference stride annotations.
2. **Spatial resolution browser**: cross-resolution accuracy curves for all 8 models and datasets; target-resolution fine-tuning by architecture family.
3. **Architecture recommender (hybrid AI)**: given a target dataset, hardware latency budget, source frame rate, and deployment context, the system performs structured data-driven analysis followed by LLM synthesis (Llama-3.3-70B via Groq), recommending the optimal model–resolution–stride operating point. Budget-constrained and unconstrained deployment paths are handled separately, with Nyquist-guided stride selection.
4. **TDS comparison**: rank any subset of the 8 datasets by temporal demand, with model-specific breakdowns and Spearman correlation display.
5. **Routing simulator**: entropy threshold sweep for any model on any dataset, with visual Pareto-frontier display.

Code availability. Evaluation scripts, model fine-tuning code (including the target-resolution fine-tuning pipeline with bicubic positional embedding interpolation), result tables, and the Streamlit dashboard source code are provided as an anonymized supplementary package for review. The public repository will be released after the review period. The package includes:

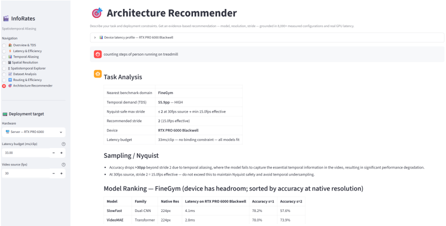
– `scripts/accv2026/`: all evaluation and retraining scripts
– `src/info_rates/`: model loading, temporal sweep, entropy routing
– `dashboard/`: Streamlit app with hybrid RAG+LLM recommender



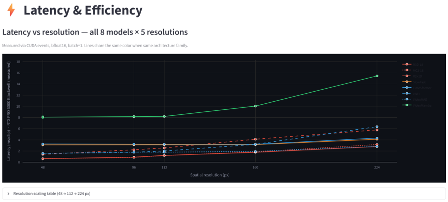
(a) Accuracy Explorer: native resolution (stride=16, cov=100%, 224px). Summary table shows top-1 across all 8 datasets.



(b) Accuracy Explorer: low resolution (stride=1, cov=75%, 48px). Retrained CNN checkpoints are used at 48px.



(c) Architecture Recommender: FineGym example. The system identifies FineGym as a high-TDS domain (55.9pp), requires stride≤2, and ranks models by accuracy under the user's latency budget.



(d) Latency & Efficiency: measured forward-pass latency (ms/clip, bfloat16, RTX PRO 6000) vs. spatial resolution for all 8 architectures. VideoMamba (green) is 4–8× slower than VideoMAE across all resolutions.

Fig. S14: Interactive dashboard screenshots from the anonymized supplementary package. The Accuracy Explorer (a,b) lets users sweep coverage, stride, and resolution interactively and view per-dataset top-1 accuracy tables. The Architecture Recommender (c) provides AI-assisted deployment guidance grounded in measured data. The Latency & Efficiency tab (d) displays GPU latency curves across all model–resolution pairs.

– `evaluations/accv2026/paper_results/`: CSV result files used in the paper.

S11 Leave-One-Architecture-Out TDS Stability

To assess whether the TDS ranking depends on the specific 8-architecture pool evaluated, we recompute TDS for all 8 datasets after removing each architecture in turn (7-architecture pools) and compare the resulting ranking to the full 8-architecture ranking via Spearman ρ.

Table S14: Leave-one-architecture-out TDS ranking stability. ρ is the Spearman correlation between the 8-dataset ranking computed with the named architecture excluded (7-architecture pool) and the full 8-architecture ranking.

Excluded architecture	Spearman ρ	p
R3D-18	1.000	$< 10^{-4}$
MC3-18	1.000	$< 10^{-4}$
R2+1D	1.000	$< 10^{-4}$
SlowFast	1.000	$< 10^{-4}$
TimeSformer	1.000	$< 10^{-4}$
ViViT	1.000	$< 10^{-4}$
VideoMAE	1.000	$< 10^{-4}$
VideoMamba	0.976	3.3×10^{-5}

For 7 of 8 leave-one-out pools, the ranking is *exactly* identical to the full-pool ranking ($\rho = 1.00$). Removing VideoMamba is the single exception ($\rho = 0.976$): it swaps the #1/#2 positions, since excluding VideoMamba raises AUTSL's average (58.3pp, from 53.0pp) and lowers FineGym's average (58.2pp, from 55.9pp)—VideoMamba is comparatively robust on AUTSL (16.0pp, below its own median drop) and comparatively fragile on FineGym (40.8pp, its worst dataset), so its presence in the pool is precisely what tips FineGym narrowly ahead of AUTSL. Ranks 3–8 (SSv2 through UCF-101) are unaffected by any single exclusion. We therefore report FineGym and AUTSL as co-equal highest-TDS domains in the main paper rather than asserting a strict order, while the rest of the ranking is robust under leave-one-architecture-out by this test.

References

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2. Shao, D., Zhao, Y., Dai, B., Loy, C.C.: FineGym: A hierarchical video dataset for fine-grained action understanding. In: CVPR. pp. 2616–2625 (2020). https://doi.org/10.1109/CVPR42600.2020.00266 10